

CYBER SECURITY · CASE STUDY REPORT · SET 2

Cyber Crime Case Studies

Banking, healthcare, ransomware and data breaches: five real incidents, what went wrong, and what they teach us.

SUBMITTED BY

NIKUNJ K

Enrollment No. 241105060064

01 COSMOS BANK

02 AIIMS DELHI

03 WANNACRY

04 BANGLADESH BANK

05 EQUIFAX

Contents

Each case follows the same six headings: what happened, attack technique, financial impact, response, recovery cost and time, and prevention. A short key lesson closes every case.

01 **Cosmos Bank Malware Attack** (Pune, India, 2018) 3
Malware on an ATM switch server drained money across many countries within hours.

02 **AIIMS Delhi Ransomware Attack** (New Delhi, India, 2022) 5
Ransomware forced one of India's largest hospitals to work on paper for weeks.

03 **WannaCry Ransomware Outbreak** (Global (150+ countries), 2017) 7
A ransomware worm locked computers in hospitals and offices worldwide in days.

04 **Bangladesh Bank SWIFT Heist** (Bangladesh / New York, 2016) 9
Fake SWIFT instructions tried to steal almost one billion US dollars.

05 **Equifax Data Breach** (USA, 2017) 11
A single unpatched system exposed the personal data of about 147 million people.

CASE 01

Cosmos Bank Malware Attack

YEAR	LOCATION	TYPE	IMPACT
2018	**Pune, India**	**Malware / ATM switch**	**About Rs. 94 crore**

In August 2018, attackers used malware to compromise the ATM switch server of Cosmos Co-operative Bank, Pune, and withdrew money across many countries within hours. It remains one of the best-known bank cyber heists in India.

Source: Indian media reports (Aug–Sep 2018) and Pune Cyber Police. Figures differ slightly between sources; verify before final submission.

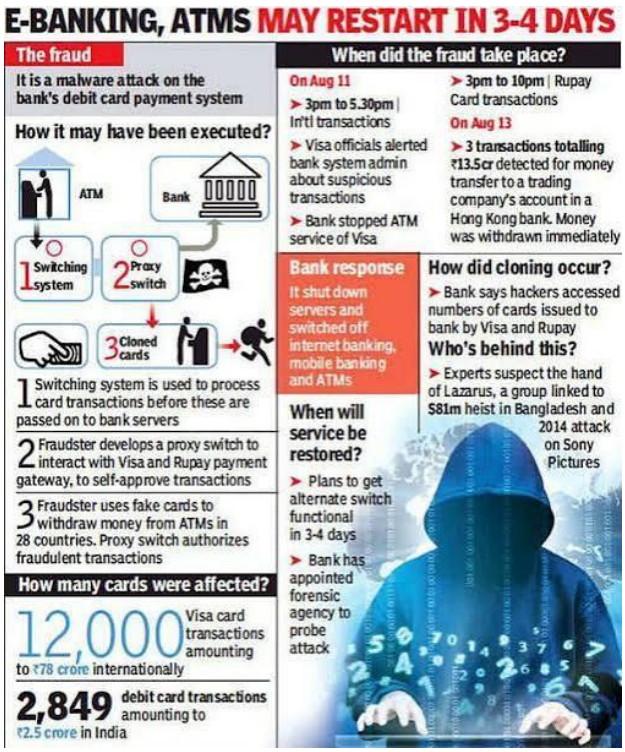


Fig. 1 How the Cosmos Bank attack was carried out (newspaper infographic). Source: publisher to be added.

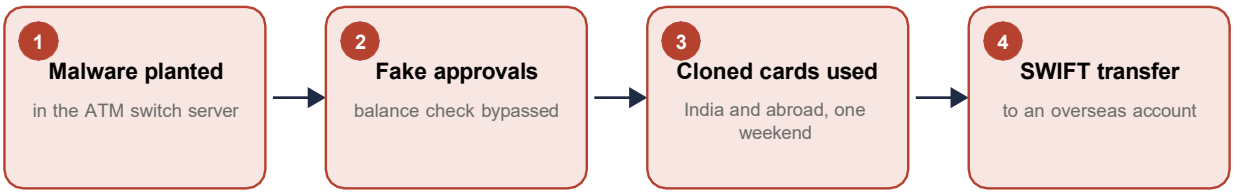


Diagram Simplified flow of the Cosmos Bank attack (schematic, not to scale).

CASE 01 · COSMOS BANK MALWARE ATTACK (CONTINUED)

1. WHAT HAPPENED

Reports say attackers first inserted malware into the bank's ATM switching system, which let them approve fraudulent withdrawal requests without the core system verifying the real account balance. Over roughly one weekend in August 2018, thousands of fraudulent transactions using cloned Visa and RuPay debit cards were carried out in India and abroad. A separate set of fraudulent SWIFT messages sent money to an overseas account. The theft came to light when the bank noticed unusual activity.

2. ATTACK TECHNIQUE

Malware on the ATM switch server (a proxy-switch attack), cloned debit cards and fraudulent SWIFT payment messages. The attackers exploited weak network segmentation and monitoring.

3. FINANCIAL IMPACT

About Rs. 94 crore was reported stolen (card withdrawals plus the SWIFT transfer). Beyond the money, customers lost trust in the bank for a period.

4. COMPANY AND AUTHORITY RESPONSE

The bank informed police and regulators, shut down the affected channels, and investigated with the Pune Cyber Police. It worked with other banks and card networks to trace and block transactions. Part of the money was reportedly traced or frozen, but most of the loss was not immediately recoverable.

5. RECOVERY COST AND TIME

The exact recovery cost is not public. Restoring systems and reconciling accounts took weeks; the legal investigation continued for years.

6. HOW IT COULD HAVE BEEN PREVENTED

Separate the ATM switch from the internet and the core banking network; monitor transaction patterns in real time (for example, many foreign withdrawals within hours); keep servers patched and protected by anti-malware; restrict and audit SWIFT access; and run regular security audits and incident drills.

> **KEY LESSON**
>
> *Banks must treat payment infrastructure as a high-value target. Network segmentation, monitoring and fast cooperation with police and card networks limit the damage.*

CASE 02

AIIMS Delhi Ransomware Attack

YEAR	LOCATION	TYPE	IMPACT
2022	**New Delhi, India**	**Ransomware**	**Not officially published**

In November 2022, the servers of the All India Institute of Medical Sciences (AIIMS), New Delhi were hit by a ransomware attack that forced one of India's largest hospitals to work manually for days.

Source: Indian media and Government of India statements (Nov–Dec 2022). Ransom demand and records affected differ between reports and are not independently verified here.



Representative image of a hacker attacking a network. Not a photo of the AIIMS incident.

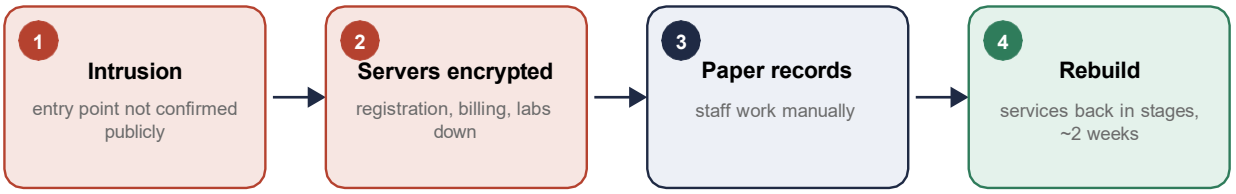


Diagram Timeline of the AIIMS Delhi ransomware incident (23 Nov 2022 onward).

CASE 02 · AIIMS DELHI RANSOMWARE ATTACK (CONTINUED)

1. WHAT HAPPENED

On 23 November 2022, hospital systems such as patient registration, appointments, billing and lab reports stopped working. The institute reported a ransomware attack on its servers, and staff recorded patient details on paper while the servers were isolated and rebuilt. Media reports said millions of patient records were at risk, and Indian agencies investigated.

2. ATTACK TECHNIQUE

Ransomware is malicious software that encrypts files and demands payment, usually in cryptocurrency. Likely entry routes for such attacks include phishing emails, unpatched systems and weak access controls; the exact entry point was not clearly confirmed in public reports.

3. FINANCIAL IMPACT

No official total loss figure is public. The impact was mainly operational: delays in patient care, about two weeks of manual working, and risk to sensitive medical data. Some early media reports mentioned a large ransom demand, which has not been confirmed officially.

4. COMPANY AND AUTHORITY RESPONSE

Servers were taken offline, the affected network was cleaned and rebuilt, and services were restored in stages. CERT-In, the National Informatics Centre and police cyber units were involved. Hospitals were also advised to improve their cyber security.

5. RECOVERY COST AND TIME

The exact recovery cost is not public. Services returned gradually over about two weeks, with full normal operation taking longer.

6. HOW IT COULD HAVE BEEN PREVENTED

Keep regular offline backups; patch software; use multi-factor authentication; segment hospital networks; train staff to spot phishing; monitor networks continuously; and prepare and test an incident-response plan.

> **KEY LESSON**
>
> *Healthcare is a critical service. Offline backups and a tested response plan are the best protection against ransomware.*

CASE 03

WannaCry Ransomware Outbreak

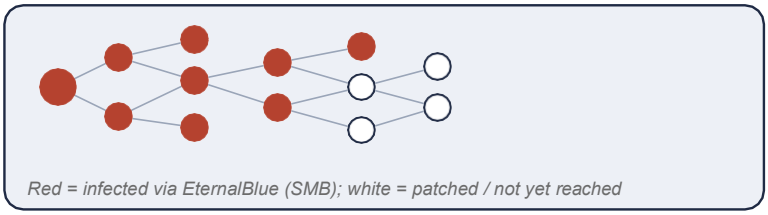
YEAR	LOCATION	TYPE	IMPACT
2017	**Global (150+ countries)**	**Ransomware worm**	**200,000+ computers**

In May 2017, the WannaCry ransomware spread across the world within days, locking computers in hospitals, companies and government offices.

Source: Microsoft, UK National Audit Office, US and UK governments and major news agencies (May 2017 onward). Estimates vary between sources.



Representative image of ATM malware 'jackpotting'. Source: securereading.com. Not from the WannaCry incident.



12 May 2017
worm starts spreading

150+
countries affected

200,000+
computers infected

Diagram Schematic of how the WannaCry worm moved between unpatched computers.

CASE 03 · WANNACRY RANSOMWARE OUTBREAK (CONTINUED)

1. WHAT HAPPENED

On 12 May 2017 the worm began spreading and infected hundreds of thousands of computers in more than 150 countries. It was especially damaging to the UK National Health Service, where many hospitals had to cancel appointments and divert patients. A security researcher found and registered a hidden “kill-switch” domain in the malware, which slowed the spread.

2. ATTACK TECHNIQUE

A ransomware worm that used the EternalBlue vulnerability in the Windows SMB service (patched by Microsoft in March 2017, bulletin MS17-010). It spread automatically between unpatched computers on the same network and encrypted their files, demanding payment in Bitcoin.

3. FINANCIAL IMPACT

Over 200,000 computers were affected. Estimates of total global losses vary widely, with some analysts suggesting up to several billion US dollars including downtime and recovery. The ransom payments themselves were comparatively small.

4. COMPANY AND AUTHORITY RESPONSE

Microsoft issued emergency patches even for unsupported Windows versions, organisations isolated infected machines and restored from backups, and governments issued security warnings. The US and UK later publicly attributed the attack to groups linked to North Korea.

5. RECOVERY COST AND TIME

Cost varied by organisation; the NHS faced large costs from cancelled appointments and IT repair. Most systems were back within days to weeks, but some took much longer.

6. HOW IT COULD HAVE BEEN PREVENTED

Apply security patches quickly; retire unsupported operating systems; block unnecessary SMB access from the internet; keep offline backups; segment networks; and train users not to open suspicious attachments.

> **KEY LESSON**
>
> *A single unpatched vulnerability can paralyse thousands of organisations. Timely patching and backups are basic but vital.*

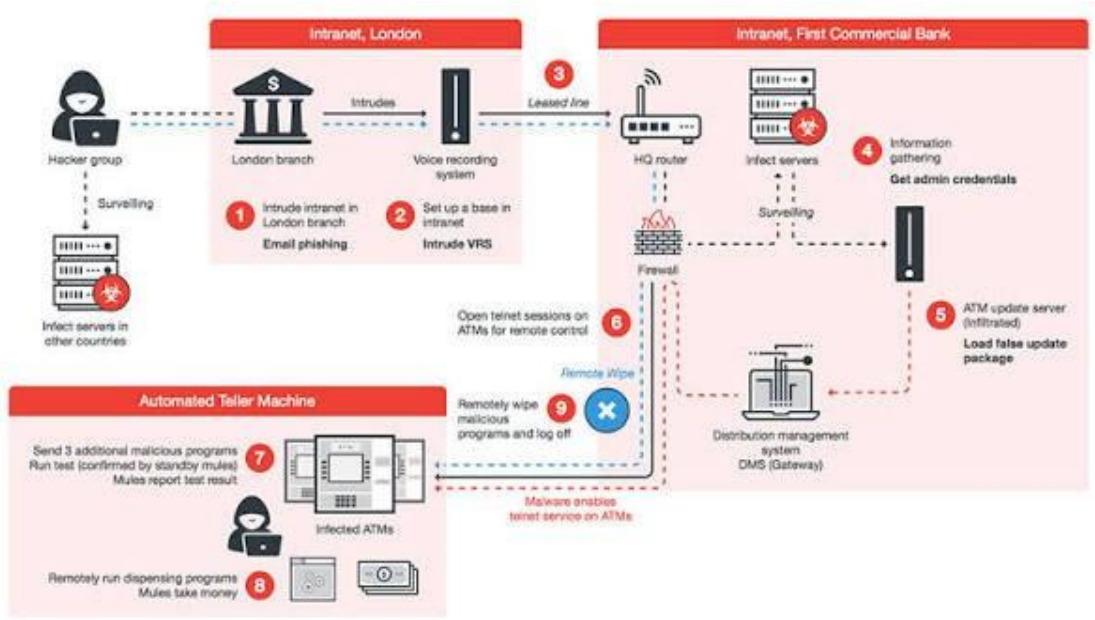
CASE 04

Bangladesh Bank SWIFT Heist

YEAR	LOCATION	TYPE	IMPACT
2016	**Bangladesh / New York**	**Fraudulent SWIFT**	**About US$81 million**

In February 2016, hackers tried to steal almost one billion US dollars from the Bangladesh Bank's account at the Federal Reserve Bank of New York by sending fake SWIFT payment instructions.

Source: International news agencies, later analysis by security firms and US authorities (2016 onward). Exact figures vary slightly by source.



Related: ATM malware heist flow at First Commercial Bank, 2016 (a different incident). Source: publisher to be added.

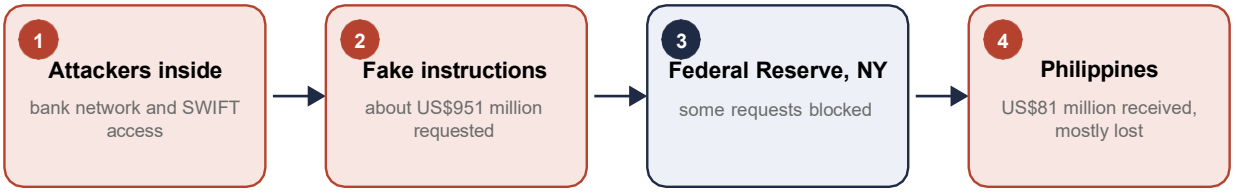


Diagram Simplified path of the fraudulent SWIFT payments (Feb 2016).

CASE 04 · BANGLADESH BANK SWIFT HEIST (CONTINUED)

1. WHAT HAPPENED

Attackers who had gained access to the bank's network sent dozens of fraudulent payment requests using the bank's SWIFT credentials. Several transfers were processed, sending money to accounts in the Philippines and Sri Lanka. Other requests were stopped, and one Sri Lanka transfer was flagged after a spelling mistake in the beneficiary name drew attention.

2. ATTACK TECHNIQUE

Malware and stolen credentials were used to access the bank's SWIFT environment, send fraudulent payment messages and hide evidence, including interfering with printed confirmations. The stolen money was then moved through accounts and casinos to launder it.

3. FINANCIAL IMPACT

The attackers tried to move about US$951 million; roughly US$81 million was successfully sent to the Philippines, and most of it was never recovered.

4. COMPANY AND AUTHORITY RESPONSE

Bangladesh Bank and international partners stopped the remaining transfers and asked the Philippines and other authorities to trace funds. Investigations involved central banks, SWIFT, police and security firms. SWIFT later introduced stronger customer security requirements. Only a small part of the stolen money was recovered.

5. RECOVERY COST AND TIME

The exact cost is not public. Senior officials of the bank resigned and years of investigation followed. A small portion of the funds was returned over time.

6. HOW IT COULD HAVE BEEN PREVENTED

Strengthen network security around payment systems; use multi-factor authentication and strict access control for SWIFT terminals; monitor and verify unusual payment instructions; and keep reconciliation and alerts running around the clock.

> **KEY LESSON**
>
> *Even highly trusted financial networks fail if the endpoints connecting to them are weak. Layered security and human verification of large transfers are essential.*

CASE 05

Equifax Data Breach

YEAR	LOCATION	TYPE	IMPACT
2017	**USA**	**Data breach**	**About 147 million people**

In 2017, the credit-reporting company Equifax suffered one of the largest data breaches in history, exposing sensitive personal information of about 147 million people.

Source: US Federal Trade Commission, US Government Accountability Office and major news agencies (2017–2019). Figures are as reported and may be updated.



PIB Fact Check graphic marking a viral headline about ATMs and UPI fees as FAKE. Shown as an example of misinformation; not related to the Equifax breach.

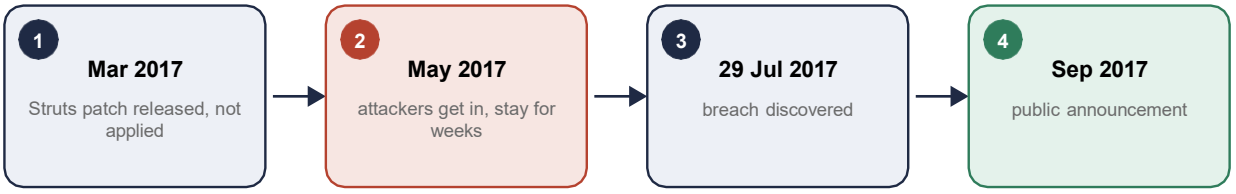


Diagram Key dates in the Equifax breach (as publicly reported).

CASE 05 · EQUIFAX DATA BREACH (CONTINUED)

1. WHAT HAPPENED

Attackers exploited a known vulnerability in a web application framework (Apache Struts) on one of Equifax's public-facing systems that had not been patched in time. They stayed inside the network for weeks and extracted data. Equifax announced the breach in September 2017, which led to widespread criticism over the delay and handling.

2. ATTACK TECHNIQUE

Exploitation of an unpatched web-application vulnerability, followed by long undetected access to databases. Weak network monitoring and expired security certificates meant the data theft was not noticed earlier.

3. FINANCIAL IMPACT

About 147 million people were affected; names, dates of birth, Social Security numbers and other data were exposed. In 2019 Equifax agreed to a settlement of up to about US$700 million with US regulators and consumers.

4. COMPANY AND AUTHORITY RESPONSE

Equifax offered free credit monitoring, replaced senior management, improved security and paid a large settlement. Regulators required stronger security programmes.

5. RECOVERY COST AND TIME

Total costs were very high, including the settlement, legal fees and security upgrades, and ran into hundreds of millions of dollars.

6. HOW IT COULD HAVE BEEN PREVENTED

Patch known vulnerabilities quickly; keep an accurate inventory of systems; monitor and encrypt sensitive data; keep security certificates up to date; and have a fast, honest breach notification process.

> **KEY LESSON**
>
> *Holding sensitive data is a major responsibility. Basic security hygiene, such as timely patching and monitoring, would have prevented a breach of this size.*